

Formal Causal Closure in Strong-Field Regimes

A Functional-Logical Derivation

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1 Scope

This document derives the strong-field limit of the Strüver framework without introducing new forces, geometric postulates, or additional ontological primitives.

Only the following primitives are admitted:

- Energy E (transition capacity)
- Mass m (bound energy density)
- Relation R
- Process Φ (finite stabilization propagation)

2 1. Weak-Field Parameter

Define the dimensionless parameter:

$$\varepsilon(r) = \frac{GM}{c_\Phi^2 r}.$$

Weak-field regime:

$$\varepsilon \ll 1.$$

Strong-field regime:

$$\varepsilon \sim 1.$$

3 2. Stabilization Propagation

Let c_Φ denote the maximal stabilization propagation rate.

Outward propagation is possible only if sufficient free transition capacity remains after internal binding.

Define outward stabilization velocity:

$$v_\Phi^{\text{out}}(r) = c_\Phi (1 - \varepsilon(r)).$$

This expresses:

- Full freedom at $\varepsilon \rightarrow 0$
- Vanishing outward capacity at $\varepsilon \rightarrow 1$

4 3. Causal Closure Condition

Causal closure occurs when:

$$v_\Phi^{\text{out}}(r) = 0.$$

Substituting:

$$\begin{aligned} 1 - \frac{GM}{c_\Phi^2 r} &= 0 \\ \Rightarrow r &= \frac{GM}{c_\Phi^2}. \end{aligned}$$

At this radius:

- Internal transitions continue.
- No outward stabilization remains admissible.
- External relational influence vanishes.

This defines the causal horizon.

5 4. Interpretation

A black hole is not a geometric object.

It is the limit state where:

bound energy density consumes all outward propagation capacity.

No additional force is introduced.

No spacetime curvature is postulated.

Only finite stabilization propagation under maximal binding.

6 5. Consistency Conditions

The derivation remains valid if:

1. c_Φ is fixed universally.
2. No additional dimensional constants are introduced.
3. Ontology does not adapt to data.
4. Layer III execution semantics are not treated as physics.

Violation of any condition invalidates closure.

7 Conclusion

Strong-field regimes are not new physical domains.

They are stabilization-dominant regimes of the same causal structure.

Causal closure arises when internal binding exhausts outward propagation capacity.

No new force is required.

Only correct functional classification.